ON-PREMISE AI VIDEO INTELLIGENCE AND WORKPLACE SAFETY SYSTEM

Technical Requirements Specification / Scope of Work

Prepared for: S Flo / Workplace Operations

Document purpose: This specification defines the functional, hardware, software, security, safety, and implementation requirements for an AI-based camera system serving entry/exit monitoring, operational visibility, safety, security, theft detection, emergency notification, and evidence-based video search.

> ***Core principle:*** *The system shall be an on-premise, evidence-based decision-support platform. It shall raise alerts and provide video evidence, but significant employment, disciplinary, safety, or security decisions must remain subject to authorized human review.*

1. Project Objective

Design, supply, install, configure, test, and hand over an AI-based multi-camera workplace video intelligence system. The system shall analyze live camera feeds and recorded video; detect configured security, safety, operational, and equipment-health events; issue immediate alerts; and answer authorized users' typed or spoken questions using verified video metadata and linked evidence clips.

• Operate primarily inside the company premises with minimal dependency on public internet.

• Continue core recording, monitoring, analytics, alerting, and local dashboard functions during internet interruption.

• Be scalable from an initial pilot of 8–16 cameras to at least 32 cameras without replacement of the core architecture.

• Provide a secure web dashboard, mobile-compatible internal access, text interface, and voice-command interface.

• Retain a complete event, investigation, acknowledgement, escalation, and audit trail.

2. Areas and Monitoring Scope

Area / Zone	Purpose	Primary analytics / alerts
Entry and exit points	Access, attendance support, security	Person/vehicle detection, line crossing, tailgating, access-control correlation, unauthorized access
Workshop / work areas	Safety, operations, incident monitoring	Presence, PPE where configured, restricted-zone entry, falls, fighting/aggression, crowding, unsafe behavior
Office and corridors	Security and occupancy	After-hours presence, intrusion, loitering, camera health
Canteen / break area	Break exceptions and safety	Occupancy, crowding, approximate long-break exceptions, fighting/fall events
Stores / warehouse	Security and inventory support	After-hours access, object removal/left behind, material movement, theft investigation
Tool room / tool crib	Tool control and evidence	Tool-board zone monitoring, removal/return events, unusual movement, custom tool recognition
Loading / parking / external yard	Vehicle and goods monitoring	Vehicle/person detection, loading activity, intrusion, safety events
Secure / restricted rooms	High-security monitoring	Unauthorized entry, door-event correlation, after-hours presence, tamper alert

Each camera must support configurable virtual lines, zones, directions, restricted areas, tool-rack zones, safety-exclusion zones, break zones, emergency routes, and loading/unloading zones.

3. Functional Requirements

3.1 Live monitoring and recording

• Central live dashboard with 1, 4, 9, 16, and 25-camera views; full-screen individual camera view; camera grouping by location.

• Continuous or policy-based recording with synchronized time across cameras, server, NVR/VMS, access-control systems, and alarm systems.

• Timeline playback with event markers, pre-event and post-event clips, snapshots, camera names, location, and confidence score.

• Role-based access control, export permissions, watermarked video export, audit logs, and retained incident evidence.

• Camera online/offline status, recording status, stream quality, disk capacity, AI service health, and network status shown on the dashboard.

3.2 Entry, exit, attendance, and break support

• Detect person crossing configured entry/exit lines and log time, direction, camera, snapshot, and clip.

• Integrate with access-control card, QR, RFID, employee badge, or approved biometric system where available.

• Flag exceptions such as tailgating, person visible without matching card event, card event without visible person, or unauthorized after-hours entry.

• Provide first-entry, last-exit, and exception reports. Camera analytics shall support—not replace—the official attendance/payroll system.

• Allow configurable break areas, shift hours, normal break durations, grace periods, exemptions, and escalation rules.

• Return evidence-linked results for queries such as: “Show employees in the canteen for more than 45 minutes yesterday.”

3.3 Incident, violence, and abnormal-event detection

• Potential fighting or aggressive physical interaction.

• Fall / person lying on the floor / possible medical emergency.

• Running in prohibited areas, crowding, abnormal gathering, loitering, and after-hours presence.

• Restricted-zone intrusion, perimeter breach, object left behind, object removed, camera tampering, blockage, defocus, repositioning, and video loss.

• Events must create a real-time alert, timestamped pre/post-event clip, snapshot, camera/location, event classification, confidence score, acknowledgement option, case status, reviewer notes, and exportable evidence package.

3.4 Theft and suspicious activity detection

• Detect and alert on unauthorized entry to stores, tool rooms, secure rooms, and restricted storage areas.

• Detect object removal from a configured shelf, rack, tool board, material area, or protected zone.

• Detect object left behind, unusual movement of selected assets, material movement through controlled areas, and after-hours access.

• Correlate video event with access-control/RFID/QR/tool-issue records where integrated.

• Provide rapid investigation queries such as: “Show all persons entering the stores after 7 PM,” “Find the video when the welding machine was removed,” or “Show access events around the time the item was last seen.”

• For reliable identification of individual small/high-value tools, provide provision for RFID, barcode/QR, BLE/UWB tags, smart cabinets, or tool-vending integration. Camera AI alone shall not be treated as definitive proof of theft.

3.5 Fire, smoke, and emergency safety alarms

The video AI solution shall complement—not replace—approved fire detection and life-safety systems. Fire alarm panels, smoke/heat detectors, manual call points, gas detectors, emergency shutdown signals, and public-address systems must remain code-compliant independent systems.

• Integrate with the Fire Alarm Control Panel (FACP) or fire alarm gateway using approved dry contact, Modbus, BACnet, OPC UA, API, or other approved interface.

• On a fire/smoke alarm, immediately show the relevant camera(s), pop up an alarm on all authorized monitoring screens, save pre-alarm and post-alarm clips, and send notifications to defined personnel.

• Where suitable camera technology and site conditions exist, support AI-assisted smoke/flame visual detection as an additional early-warning layer.

• Support integration of gas leak, hydrogen leak, high temperature, electrical panel alarm, emergency stop, intrusion alarm, panic button, door forced-open, water leak, and other safety alarms.

• Create a configured emergency escalation workflow: alert → acknowledgement → escalation if unacknowledged → incident case → closure after authorized review.

• Provide alarm priority levels: Critical, High, Medium, Low. Fire, gas, hydrogen, serious fall, violence, and perimeter intrusion shall normally be Critical or High based on site policy.

• Provide emergency notifications through local dashboard pop-up, audible alarm, beacon/siren relay where approved, email/SMS/WhatsApp/secure messaging gateway, and optional mobile push notification, subject to company policy and local regulations.

4. Generative AI Query and Voice Interface

• Accept written English commands and spoken English commands; optional Tamil and mixed English/Tamil support shall be included as a phase-2 requirement.

• Convert voice to text, interpret the request, retrieve relevant indexed video metadata/events, and return evidence-grounded answers.

• Answer with video timestamps, camera/location, snapshots, confidence, and clickable clips—not unsupported conclusions.

• Allow follow-up questions, summary generation, daily incident reports, and voice/text response.

• Maintain an audit log of user prompts, search results, retrieved evidence, generated response, and actions taken.

• Clearly state uncertainty, for example: “Possible object removal detected; human review is required.”

> *Example command: “Show all after-hours entries into Stores 2 last Friday and identify unmatched access-card events.”*

Required response: A list of matched video events with camera, timestamp, access-control correlation result, confidence, snapshot, and link to each video clip. The system must never claim that a person “stole” an item unless an authorized human investigator records that conclusion.

5. Alerts, Notification, and Action Workflow

Alarm class	Examples	Required immediate action
Critical	Fire alarm, gas/hydrogen alarm, smoke/flame alert, serious fall, active fight/violence, major intrusion	Instant dashboard pop-up, audible alert, relevant live view, pre/post clips, notify security/safety manager and escalation contacts; log acknowledgement
High	Theft/suspicious removal, unauthorized secure-room access, camera tamper, perimeter breach, door forced-open	Immediate alert and clip; notify designated security/operations users; escalate if not acknowledged within configured period
Medium	Long break exception, loitering, crowding, object left behind, repeated access mismatch	Dashboard/task notification; include in review report; escalation by rule
Low	Camera quality degradation, storage threshold warning, non-critical analytics exception	Dashboard and maintenance notification; ticket creation where integrated

• Every alert must record: event ID, alarm class, camera/location, time, related sensor/alarm source, snapshot, clip, AI confidence, assigned responder, acknowledgement time, action notes, final disposition, and closure approval.

• If a Critical alarm is not acknowledged within a configurable period, the system must escalate to the next-level contact automatically.

• Provide a daily dashboard for open alarms, unresolved cases, repeat events, camera faults, storage alarms, and notification delivery status.

• Alarm workflow shall be configurable by area, shift, business hours, event type, and authorized recipients.

6. Camera and Field Device Specification

Parameter	Minimum	Recommended
Camera type	IP network camera	IP AI camera with edge analytics metadata
Resolution	4 MP	4K / 8 MP for gates, stores, tool zones, and wide areas
Frame rate	20 fps	25–30 fps for gates, incidents, safety-critical zones
Compression	H.264	H.265 / H.265+ with H.264 fallback
Low light	IR night vision	True WDR, IR, low-light color where justified
WDR	Preferred	120 dB or greater at entrances
Lens	Fixed allowed	Motorized varifocal for gates/stores/corridors
Network	Ethernet	PoE/PoE+, managed switch environment
Interoperability	ONVIF compatible	ONVIF Profile T and analytics/metadata support
Outdoor protection	As required	IP66/IP67 and IK10 where exposed/vandal risk
Cybersecurity	Unique passwords	HTTPS, signed firmware, secure update control, disabled defaults

The supplier shall perform a site survey and submit field-of-view drawings, mounting height, lighting assessment, blind-spot assessment, coverage overlap, cable routes, PoE loading, grounding/surge protection, and rule-zone overlay for every camera.

7. On-Premise Architecture and Network

Required architecture: IP cameras → managed PoE switches → segregated CCTV/AI VLAN → VMS/NVR recording and storage → GPU AI inference server → event metadata database → generative-AI retrieval service → secure local web dashboard and authorized remote VPN access.

• Separate CCTV/AI network VLAN from office IT network; do not expose cameras directly to the public internet.

• Gigabit Ethernet minimum to edge switches; 10 GbE uplink between core switch, server, and storage for systems above approximately 16 high-resolution cameras.

• Use firewall rules, VPN, MFA, HTTPS/TLS, role-based user permissions, and centralized time synchronization using NTP.

• Provide UPS backup for core switch, PoE switches, AI server, VMS/NVR, storage, firewall, and alarm gateway as applicable.

• Provide local operation during internet outage; internet should be used only for authorized remote access, notifications, backups, or software updates.

8. Server, GPU, and Storage Requirements

8.1 Initial 8–16 camera deployment

Component	Recommended production specification
CPU	Intel Xeon W/Xeon Scalable or AMD EPYC/Ryzen 9 class; minimum 16 physical cores
GPU	NVIDIA RTX 6000 Ada 48 GB or NVIDIA L40S 48 GB preferred for 24/7 enterprise operation
Pilot GPU minimum	NVIDIA RTX 4090 24 GB may be used for pilot/small deployment subject to thermal, warranty, and support requirements
RAM	128 GB ECC preferred; 64 GB minimum
System storage	2 × 1.92 TB enterprise NVMe SSD in RAID 1
AI/event storage	2–4 TB enterprise NVMe SSD for models, metadata, clips, database
Network	Dual 10 GbE preferred
Power	Redundant server PSU preferred; correctly sized UPS
Operating system	Ubuntu Server LTS preferred for AI stack; selected VMS components may use Windows Server

8.2 16–32 camera deployment

• AMD EPYC or Intel Xeon Scalable, 24–32 physical cores.

• One 48 GB enterprise GPU (NVIDIA L40S / RTX 6000 Ada) with chassis, power, cooling, PCIe lanes, and software licensing provision for a second GPU.

• 128–256 GB ECC RAM, rack-mounted chassis, redundant PSU, dedicated RAID recording storage, and 10/25 GbE network connectivity.

• Final sizing must be based on the number of streams, resolution, bitrate, fps, models running per stream, retention period, and benchmark test in the actual environment.

8.3 Video storage

The supplier must calculate storage based on actual bitrate and retention. Use the planning formula: Storage/day = Number of cameras × average bitrate per camera × 86,400 ÷ 8. For illustration, 16 cameras at 4 Mbps average bitrate generate approximately 691 GB/day and approximately 20.7 TB for 30 days before RAID, overhead, evidence clips, and reserve capacity.

Deployment	30-day target usable storage	Recommended approach
8 cameras	12–20 TB	Enterprise surveillance/nearline disks; RAID 6 or RAID 10; separate NVMe metadata volume
16 cameras	24–40 TB	Dedicated RAID storage; reserve capacity for clips, database, and future cameras
32 cameras	48–80 TB	Separate recording/storage node or scalable NAS/SAN; 10/25 GbE where required

9. Camera Fault and System Health Monitoring

• Immediately notify authorized users if any camera becomes offline, loses video, stops recording, is blocked, covered, moved, defocused, excessively dirty, tampered with, or experiences repeated stream failure.

• Monitor camera power/PoE status, network connectivity, packet loss where available, stream bitrate, frame rate, recording continuity, disk health, server CPU/GPU/RAM utilization, temperature, fan health, and UPS status.

• Generate storage-capacity warnings at configurable thresholds, for example 70%, 80%, 90%, and critical capacity.

• Provide maintenance dashboard with camera name, fault type, first detected time, duration, assigned technician, corrective action, and closure status.

• Send immediate notification for camera fault in critical zones such as entry gate, secure stores, fire-sensitive areas, tool room, and perimeter.

• Provide monthly camera-health, recording-health, storage-health, and incident-response reports.

10. Safety, Privacy, and Security Controls

• Comply with applicable local laws, client rules, workplace policy, and data-protection obligations. Obtain legal/HR review before biometric recognition or audio recording.

• No cameras in toilets, changing rooms, prayer/private rooms, or any prohibited/private area.

• Audio recording disabled by default unless explicitly approved, legally compliant, and clearly notified.

• Face recognition disabled by default and only enabled after written approval, defined purpose, lawful basis, accuracy validation, retention rules, and access controls.

• No automatic disciplinary action, salary deduction, termination decision, or accusation of theft based solely on AI output.

• Human reviewer must confirm all high-impact security, safety, HR, and theft-related events before final action.

• Use MFA for administrators and remote users; unique camera credentials; encryption in transit and at rest; secure backups; patch and firmware-management process.

• Maintain immutable or protected audit records for logins, video playback, export, deletions, configuration changes, alert acknowledgement, AI prompts, and case actions.

11. Implementation Phases

Phase	Scope and deliverables
1. Survey and design	Site survey, camera layout, field-of-view drawing, network design, BOQ, server/GPU/storage sizing, alarm integration design, privacy/security plan, implementation schedule
2. Core deployment	Install cameras, PoE/network, UPS, server/storage, VMS/NVR, recording, camera health dashboard, basic analytics
3. Safety and security alarms	Configure fire-panel/sensor integration, critical alarm workflow, notifications, escalation, incident case management, theft and restricted-access rules
4. GenAI and search	Video metadata index, historical search, typed conversational AI, evidence-based responses, reports, voice interface
5. Advanced/custom models	PPE, named tool/asset recognition, Tamil support, RFID/QR/BLE/UWB integration, HRMS/ERP/tool-management integration
6. Testing and handover	Site acceptance test, training, documentation, benchmark report, limitations register, support handover

12. Acceptance Criteria

Requirement	Target
Live-view latency on LAN	Normally less than 2–5 seconds, subject to selected camera/network configuration
Standard AI event alert	Normally less than 10 seconds from detected event
Camera/video loss notification	Within 60 seconds or tighter where technically feasible
Historical indexed search	Normally within 30 seconds for standard queries across 30-day indexed footage, subject to query complexity
Playback start on LAN	Normally within 5 seconds for standard clips
System availability target	99.5% or better excluding planned maintenance
Accuracy reporting	Supplier must measure false positives/false negatives separately for each use case; generic “99% accuracy” is not acceptable
Evidence integrity	Exported clips watermarked and auditable; event record includes source camera/time and reviewer actions
Critical-alarm workflow	Alert, acknowledgement, escalation, notification delivery log, incident closure process demonstrated during SAT

13. Supplier Deliverables

• Complete system architecture, BOQ, make/model details, licenses, warranty, and country of origin.

• Camera layout, cable routing, network topology, IP plan, rack layout, power/UPS calculations, storage calculation, GPU sizing calculation.

• AI capability matrix stating: standard feature, configuration-only feature, custom-training feature, limitation, expected accuracy, and test method.

• Fire/safety/alarm integration drawings, cause-and-effect matrix, escalation matrix, and notification test record.

• Cybersecurity hardening checklist, backup/disaster-recovery procedure, user roles matrix, retention policy, and privacy controls.

• Site Acceptance Test report, training material, administrator/user manuals, API documentation, source-code/licensing clarification, and annual maintenance proposal.

• Documented limitations and operating assumptions for fight/fall/fire/smoke/theft/tool-recognition analytics.

14. Recommended Pilot

Start with an 8-camera pilot before full deployment: 2 cameras at entry/exit, 2 in workshop/work area, 1 in canteen, 2 in stores/tool room, and 1 at loading/outdoor area. Use 30-day retention, on-premise GPU AI server, and the following initial use cases: entry/exit events, restricted-zone intrusion, after-hours presence, camera tampering/faults, fire-panel alarm integration, selected safety alarms, long-break exceptions, object removal from designated tool-storage zones, theft investigation search, and text-based historical video search.

After 30–60 days, review alert quality, false alarms, camera angle/lighting, storage consumption, GPU usage, notification response time, required new rules, and whether RFID/QR/other asset technologies are needed before expanding to advanced named-tool or biometric features.

15. Instruction to Technical Team

We require an on-premise AI video intelligence and workplace safety system using multiple IP cameras at entry/exit points, work areas, offices, canteen, stores, tool room, loading areas, external yard, and secure zones. The system shall record and analyze live and stored video; support employee entry/exit and break-time exception monitoring; detect security, theft, safety, and operational events; integrate critical fire and safety alarms; notify and escalate incidents immediately; detect camera faults; and provide auditable evidence clips.

The system shall include a Generative AI assistant that accepts typed and spoken commands, searches indexed video/event data, and gives evidence-based answers with timestamps, camera names, snapshots, confidence, and clickable video clips. It must be designed for local GPU-based processing, local storage, minimal internet dependency, secure role-based access, audit logs, and future integration with access control, RFID, HRMS, ERP, fire alarm, gas detection, emergency shutdown, and tool-management systems.

Please submit: site survey, camera layout, hardware/software BOQ, GPU/RAM/storage sizing, network architecture, safety-alarm integration design, AI use-case matrix, privacy/cybersecurity plan, implementation programme, support model, and acceptance-test plan.